

IMO 2026 — Autonomous Model Comparison

Independently graded results across four autonomous runs

Compiled July 18, 2026 · grading: six independent verifier agents, IMO 0–7 scale, claims checked symbolically/numerically

Graded scores (not self-reported)

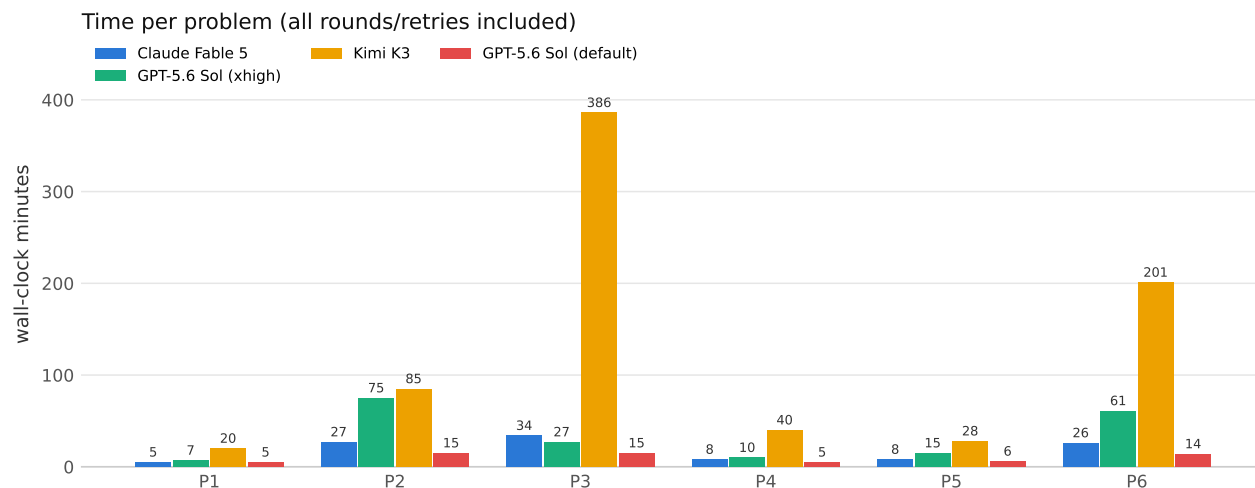
Run	P1	P2	P3	P4	P5	P6	Total
Claude Fable 5	7	7	7	7	7	7	42/42
GPT-5.6 Sol (xhigh effort)	7	4	7	7	7	7	39/42
Kimi K3	7	7	5	7	7	3	36/42
GPT-5.6 Sol (default effort)	7	4	2	7	7	1	28/42

Every run except Opus 4.8 (honest “partial” on P3) self-reported all its attempted problems as solved; the grades above reflect independent verification. Notable deductions: Kimi K3’s P3 game-value lemma makes a claim refuted by exact minimax and its P6 finiteness crux is proved only for finite transversals; GPT-5.6 Sol (lw)’s P6 central lemma is refuted by an explicit machine-verified counterexample and its P3 rests on an unproven sketch; Opus 4.5 repeatedly asserts cruxes (“SymPy confirms”) without proof.

Time and cost

Run	Graded score	Total time	Cost (USD)	Output tokens
Claude Fable 5	42/42	1.8 h	\$38.83	371k
GPT-5.6 Sol (xhigh effort)	39/42	3.3 h	~\$11.48	174k
Kimi K3	36/42	12.7 h	~\$28.20	1,401k
GPT-5.6 Sol (default effort)	28/42	1.0 h	~\$4.04	80k

Claude Fable 5 cost as metered by the Anthropic API. Kimi K3 estimated from logged tokens at list pricing (\$0.30/M cache-hit, \$3/M miss, \$15/M output, 93% hit assumption); GPT-5.6 Sol from logged tokens at OpenRouter list pricing (\$5/M in, \$30/M out). Kimi K3 time includes all retry rounds (P3 took four).



Findings

- **Claude Fable 5: verified 42/42**, and the fastest run (1.8 h) — full rigor in a bare single-context loop with no reviewer, no orchestration.
- **Reasoning effort buys rigor**: GPT-5.6 Sol jumped 28→39/42 going from default to **xhigh** effort in the identical harness — P3 and P6 became verified 7s, and its P2 failure turned into an honestly-declared partial instead of an invalid proof.
- **Kimi K3’s 36/42 is gold-medal-range work** (typical human gold cutoffs are far lower), earned over 12.7 h including three failed rounds on P3 whose in-context work was lost to time caps before checkpoint countermeasures were added to the harness.
- **Self-reports inflate**: graders confirmed only some of the “solved” claims as complete proofs; every deduction is documented with a named, checkable defect. All stated answers to the compute-and-prove problems agree across all runs.

Caveats

- Graders are Claude-based agents; despite adversarial instructions, machine-checked counterexamples, and symbolic verification, they are not human medalists.
- All four runs used the identical harness (same system prompt, tools, time caps), so they are directly comparable. Reasoning settings: Claude Fable 5 at its API default effort (**high**, always-on adaptive thinking); Kimi K3 at its API default; GPT-5.6 Sol at default (**medium**) and **xhigh** — the one controlled difference between its two runs.
- Full per-turn audit trails for every run, plus per-problem grader verdicts with named defects, accompany this document.